TECHGARDEN
FRANKFURT

TECHGARDEN #1 · 12 NOVEMBER 2026

Comparing AI models.

Open and closed. European and non-European.
What can they really do?

Narges Hadji-Hosseini · techgarden-frankfurt.de

Hosted by HANF Advanced Fire Engineering

TECHGARDEN
FRANKFURT

WELCOME TO TECHGARDEN #1

Tonight we build together.

6 pm	Then	~7:30 pm	~8:30 pm	9 pm
Welcome and a short talk	Build in small groups	Pizza	Closing round	End



Hosted by our sponsor
Hanf Advanced Fire Engineering

Pizza sponsored by **TechGarden Frankfurt**

THE HOOK

A post went viral on LinkedIn

“Revolut banned OpenAI and Anthropic. For over 2,000 developers.”

“The replacement: self-hosted open-weight models from China.”

“AI costs down 90 percent. Teams shipped 70 percent faster.”

“No API ping to California.”

Claims paraphrased from widely shared LinkedIn posts, November 2026

THE SOURCE

What the talk actually said

The post says	The talk says
Banned OpenAI and Anthropic	“We keep using Codex, Claude Code … but we keep them as exception.” Open models are the default, not the only option.
No data to the US	Weights run “on a US-based service”. The US counts as an acceptable location for them.
90 % lower costs	Roughly right: the invoice is “one-tenth” of what they spent before.
70 % faster because of the model	Lead time improved, but token budgets were lifted and tools got better at the same time.

Source: David Abitbol (Revolut), “Shipping like a startup, regulated like a bank”, AI Engineer NYC, 13 October 2026

Lots of noise.

Tonight we check for ourselves.

THE MAP

Four fields, many models

	Open · run it yourself	Closed · use the provider
European	Mistral (open models), Apertus, Teuken	Mistral (large models), Aleph Alpha
Non-European	Llama, Gemma, gpt-oss, Qwen, DeepSeek, GLM, Kimi	GPT, Claude, Gemini
	+ control, runs anywhere − you operate it	+ strongest, easiest to start − data goes to the provider

SOVEREIGNTY

Sovereignty has layers

Data	Where is it processed and stored?
Operation	Who runs the model, on whose servers?
Model	Who trained it, and who ships the next version?
Hardware	Whose chips does it run on?

Revolut: own operation and an open model, but US hosting and NVIDIA chips.

Local is not the same as trusted: the Bundeswehr is phasing out Chinese drones and cars, US agencies banned DeepSeek. A local model can’t phone home, but nobody can fully check what it was trained to do.

A WORD ABOUT THE WORD

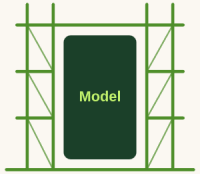
Harness or scaffolding?



THE USUAL WORD

Harness

A horse harness steers raw power with reins. But nobody pulls a model by the reins.



THE BETTER PICTURE

Scaffolding

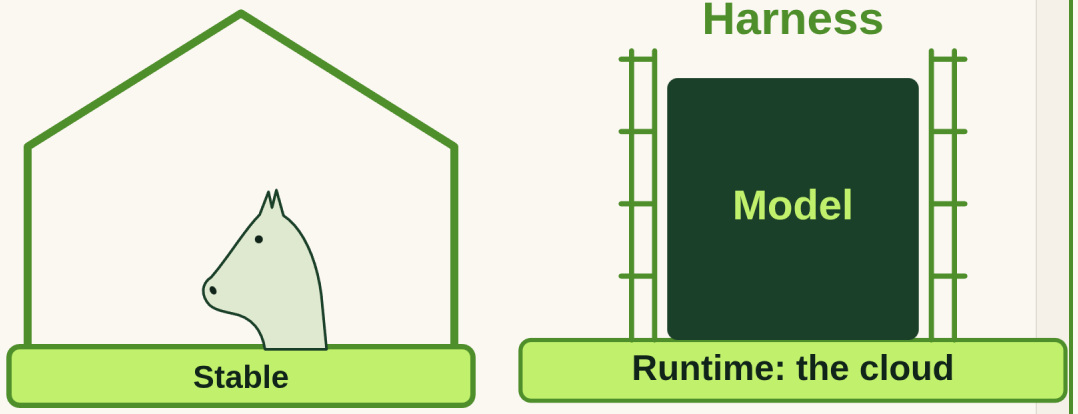
A structure built around the model: tools, memory, instructions, steps. You can build on it.

Tonight we say **harness**, because everyone does. **Think scaffolding.**

THE OVERLOOKED LAYER

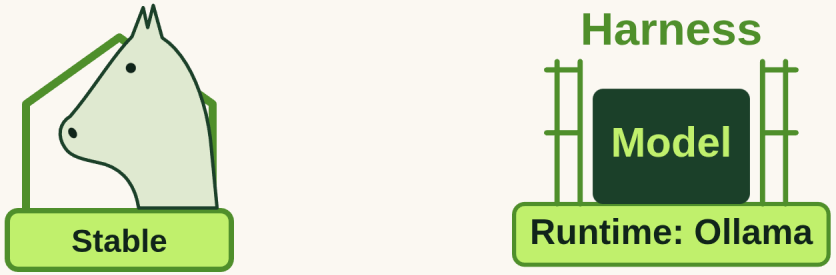
Room to move: the runtime

CLAUDE CODE · CODEX



Harness on your laptop, the rest in the cloud. **Set up for you.**

QWEN CODE + OLLAMA



Everything on your laptop. **Context, memory, speed: up to you.**

Tested yesterday: Ollama’s default context of 4,096 tokens cut off Qwen Code’s instructions, and the model looked dumb. With 16,384 it worked, using 7.9 GB of RAM.

BEFORE WE START

A model is not a harness

Model

The trained network that writes the answer.

Claude · GPT · Gemini · Mistral · Qwen · Llama

Harness

Everything around the model: tools, files, memory, instructions, and the loop that lets it work in steps. Also called scaffolding or environment.

Chat · Claude Code · Codex · Qwen Code · Cursor · Copilot

Think scaffolding: **the model does the work, the structure around it decides what it can reach.** Same model, different harness, different result. Always note both.

Even a plain chat is a harness. For a fair model comparison, use the same harness for every model, e.g. the OpenRouter chat.

TONIGHT

Same task, several models

1. Pick a task, or bring your own
2. Run it on two or more models
3. Note the results in results.csv
4. Open a pull request

1 · INDUSTRY TASKS · HANF ADVANCED FIRE ENGINEERING GMBH

Analyze a BIM model · Sort project emails · Automated model checking with IDS

Start here if you can. Mr Hanf, the founder, is here tonight to answer questions. Strong solutions may lead to a follow-up.

2 · General tasks: summarize a German technical text · read a table from a PDF · say “I don’t know” · write code

github.com/techgarden-frankfurt/ai-model-comparison

Local AI or AI-built software?

WAY A

AI runs locally

Real data → **Local model** → Result

A model on your own hardware reads the confidential data. Nothing leaves the house.

Fits fuzzy work: free text, emails.
Costs: hardware, lower quality, results vary.

WAY B

AI builds, software runs

Sample data → **AI writes code** → Program

Real data → **Program** → Result

Fits rules: IDS checks, BIM analysis.
Gains: no AI sees the real data, same result every time.

Tonight: IDS and BIM → Way B · Emails → Way A, or rules plus a local model.

RECOMMENDED SETUPS

Which model, which harness

1 · INDUSTRY TASKS

Solve the task. Respect the data rules.

Cloud: Claude Code or Codex

Data stays on the laptop: Ollama with a small model, e.g. qwen3:8b

Public sample models and made-up emails only. Nothing confidential into any model. Note where the data went: US, EU or local.

2 · GENERAL TASKS

Same prompt, several setups.

Claude	Claude Code or chat	closed · US
GPT	Codex or ChatGPT	closed · US
Qwen	Qwen Code	open · CN
Mistral	Le Chat	EU
Local	Ollama, small model	laptop

Good to know: open weights don’t mean a free service. Qwen Code in the cloud sends your data to Alibaba, a Chinese provider: check the region. Chat rejects .ifc? Rename a copy to .txt.

For the curious: run Qwen inside Claude Code. Same model, different harness.

GROUND RULES

Six rules for the evening

- ✓ Public data only
- ✓ Note model, version, harness, date
- ✓ Nobody has to be a pro
- ✓ No API keys in the repository
- ✓ Bad results count too
- ✓ Ask when you’re stuck

FOR THE CLOSING ROUND

Will everything run locally in three years?

More and more, yes

Laptop models today are about as good as the frontier one or two years ago.

Models get smaller and more efficient, laptops get more and faster memory.

Everyday tasks will often stay on the device.

Not everything

Frontier models like Opus or Fable are not released as weights, and far too big.

The frontier keeps moving, and training always needs data centres.

Cheaper AI means more use: more agents, longer tasks, video.

Open question: are we building too many data centres? Some compare it to the fibre boom around 2000.

TECHGARDEN
FRANKFURT

Come and build something.

Every other Thursday, 6–9 pm, in Frankfurt. Bring your laptop and a project.

Join the next meetup ↗ **techgarden-frankfurt.de ↗**



Thanks to our sponsor for hosting tonight
Hanf Advanced Fire Engineering

Pizza sponsored by **TechGarden Frankfurt**

BUILD. CODE. SHIP.

BACKUP

For questions.

Basics, key terms, RAG, memory, hardware and NVIDIA.

THE BASICS

Three things to know

What is a language model?

A program trained on huge amounts of text. It reads text and writes text.

Closed or open

Closed: you use it through the provider. **Open:** you download the weights and run it yourself.

How to try one

In a browser chat, through an API, or locally with Ollama or LM Studio.

KEY TERMS

Ten words you'll hear tonight

Generative AI · AI that creates new content: text, images, code.

LLM · large language model, a generative model for text.

Token · the unit a model reads and writes, roughly ¾ of a word. Prices and limits count tokens.

Context window · how much text a model can take into account at once.

Hallucination · a confident answer that is made up.

Open weights · the trained model files are published, so you can run it yourself.

Quantization · storing weights with fewer bits, e.g. 4-bit, so a model fits smaller hardware.

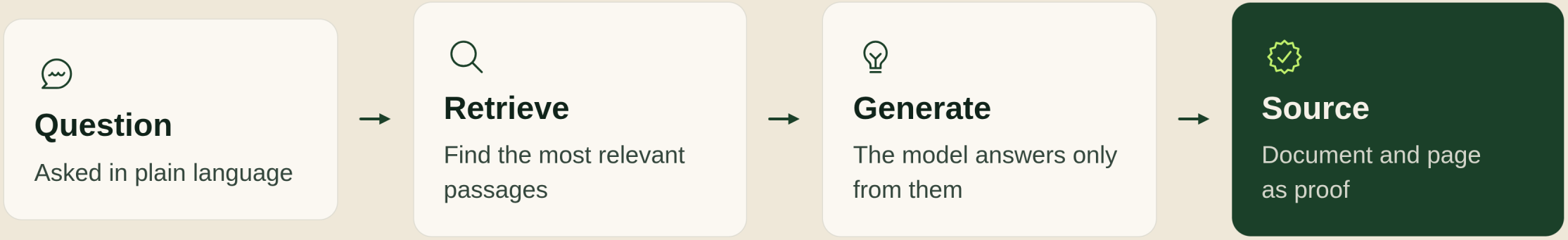
Harness · everything around a model: tools, files, memory, instructions, the loop. Also: scaffolding.

Agent · a model that plans and uses tools over several steps to reach a goal.

MCP · Model Context Protocol, a standard way to connect models to tools and data.

RAG · RETRIEVAL-AUGMENTED GENERATION

Answers from your own documents



Why it matters: company knowledge without retraining the model, and answers you can check.

Watch out: if the search misses the right passage, the answer misses it too.

RUNNING MODELS LOCALLY

Memory decides

≈ 0.6 GB

per billion parameters, at 4-bit quantization, plus room for the context

8B ≈ 5 GB · 14B ≈ 9 GB · 70B ≈ 40 GB

For every word it writes, the model is read from memory once. Two things matter:

1. Size: does the model fit into the memory the GPU can use?

2. Bandwidth: how fast can that memory be read?

Which hardware for which goal

Goal	Model size	Typical hardware
Try it out	up to ~14B	A laptop with 32 GB RAM, a MacBook with 16–24 GB
Good everyday quality	~30–70B	Mac with 64–128 GB, AMD Ryzen AI Max with 128 GB, NVIDIA DGX Spark
Very fast, mid-size	up to ~30B	PC with a strong NVIDIA GPU, 24–32 GB VRAM
Full-size GLM, DeepSeek	hundreds of B	Servers with many data-centre GPUs (Revolut: 56 × B200)

Apple and AMD unified memory: fits big models. **NVIDIA GPUs:** fastest, as long as the model fits.

THE BOTTOM LAYER

Why almost everything runs on NVIDIA

Hardware fits the job

Thousands of cores, Tensor Cores for matrix maths, very fast memory.

CUDA since 2007

PyTorch, vLLM and FlashAttention land there first. A network effect.

Scaling out

NVLink and data-centre networking make many GPUs act as one.

Catching up: AMD ROCm · Apple MLX · Google TPU